

GeoCVS

Measuring the Critical View of Safety from exemplar-prompted masks and explicit geometric predicates, and quantifying how few masks the measurement requires.

Key Problem

Bile duct injury during laparoscopic cholecystectomy is a rare but serious complication whose dominant cause is a perceptual error: the surgeon misidentifies the common bile duct as the cystic duct. The Critical View of Safety prevents this error through three verifiable conditions (two tubular structures entering the gallbladder, dissection of the lower third of the gallbladder from the liver bed, and clearance of the hepatocystic triangle), and its value derives from being a checklist of measurements a surgeon can contest. Existing systems instead produce the decision with a head trained on Critical View labels, whether over an intermediate segmentation [Mascagni et al., 2022] or latent scene-graph relations [Murali et al., 2022], so the geometry is only an internal feature and a scalar such as 0.62 offers no basis a surgeon can interrogate. Whether a measured alternative is feasible depends on one prerequisite: whether frozen segmentation is accurate enough on the fine tubular anatomy the Critical View depends upon. This project measures that prerequisite rather than assuming it.

Motivation

The project comprises two parts. **Part A** measures the mask-efficiency of frozen SAM 3 [Carion et al., 2025] on the six criterion-relevant concepts (gallbladder, cystic duct and artery, liver bed, fat, instrument): how few image exemplars per concept are required before zero-shot segmentation matches a supervised segmenter, with SurgicalSAM [Yue et al., 2023] and MedSAM [Ma et al., 2023] as references, expecting near-parity for large structures and a residual deficit for the fine cystic duct and artery. **Part B** interprets the three criteria as geometric statements (an instance count, a surface-separation measurement, and a region-occupancy measurement) and computes each as an explicit thresholded predicate over the Part A masks and Depth Anything 3 geometry [Lin et al., 2025], calibrating three scalar thresholds on about fifty frames rather than training a head on eleven thousand, with conformal abstention under ambiguity [Angelopoulos and Bates, 2021]. Because the thresholds are physical rather than fitted to rater consensus, the method should degrade less under domain shift, which the SAGES CVS Challenge [Alapatt et al., 2025] evaluates directly, and being training-free it can enter that hidden-test leaderboard. The defensible claim is near-parity in domain, with advantages in auditability, label efficiency, and transfer; because the available surgical depth evidence concerns a different model, the interface depth is validated against SCARED stereo ground truth [Allan et al., 2021] rather than assumed.

Datasets

- **SAGES CVS Challenge** [Alapatt et al., 2025] (primary evaluation): 1,000 cholecystectomy videos, 20 raters, a hidden Docker-scored test set, and three subchallenges (achievement, uncertainty, robustness).
- **Endoscapes-CVS** [Murali et al., 2023] (mask source and Part A substrate): 201 videos, 11,090 Critical-View-annotated frames from 3 raters, 422 mask-annotated frames; the challenge provides no masks.
- **CholecSeg8k** [Hong et al., 2020] and **Cholec80** [Twinanda et al., 2016]: additional segmentation and transfer data.
- **ATLAS-120k** [de Jong et al., 2026]: multi-procedure anatomy, a stress test of mask quality under procedural diversity.

Most Relevant Papers

- **Alapatt et al. (2025), The SAGES Critical View of Safety Challenge.** The primary evaluation benchmark; its subchallenges instantiate the accuracy, uncertainty, and robustness claims. [Alapatt et al., 2025]
- **Murali et al. (2023), Latent Graph Representations for Critical View of Safety Assessment.** The most directly comparable prior method; it also encodes geometry, but as latent relations decided by a trained head. [Murali et al., 2022]
- **Mascagni et al. (2022), Artificial Intelligence for Surgical Safety (Annals of Surgery).** The segmentation-then-assessment reference for automatic Critical View assessment. [Mascagni et al., 2022]
- **Carion et al. (2025), SAM 3: Segment Anything with Concepts.** Provides the exemplar-prompted instance masks studied in Part A. [Carion et al., 2025]
- **Yue et al. (2023), SurgicalSAM.** A trained, class-promptable segmenter; with MedSAM, the reference against which the Part A scaling curve is measured. [Yue et al., 2023]
- **Lin et al. (2025), Depth Anything 3.** Provides the dense geometry for the depth-based separation and adjacency predicates. [Lin et al., 2025]
- **Angelopoulos and Bates (2021), A Gentle Introduction to Conformal Prediction.** The distribution-free method for calibrated abstention. [Angelopoulos and Bates, 2021]

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